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Nomenclature

Greek Symbols

π $\simeq 3.14\dots$

Acronyms / Abbreviations

AMO Atlantic Multidecadal Oscillation

BSISO Boreal Summer Intraseasonal Oscillation

ENSO El Niño–Southern Oscillation

MJO Madden-Julian Oscillation

NAO North Atlantic Oscillation

PC Principal Component

PDO Pacific Decadal Oscillation

VC Vector of Coefficients

Theoretical Foundations

1.1 | Principal Component Analysis

Principal Component Analysis (PCA) is a statistical technique used to reduce the dimensionality of a dataset while preserving most of the variance. This is achieved by identifying the principal axes, which are the directions in which the data varies the most. These directions correspond to the eigenvectors of the covariance matrix of the data, with the associated eigenvalues representing the magnitude of variance along these directions. PCA is particularly useful for dimensionality reduction, data compression, feature extraction and visualization of high-dimensional datasets.

The origins of PCA can be traced back to Pearson [55] in 1901 and Hotelling [37] in 1936. It gained significant traction in meteorology through the work of Lorenz [49] in 1956, who introduced the technique under the term *empirical orthogonanal function* (EOF) analysis¹. The method has since been extensively applied in meterology and climate science to identify dominant modes of variability in the atmosphere and oceans, as discussed in numerous textbooks [e.g. 21, 31, 40, 56, 64, 72].

In this section, we derive PCA from the maximum variance formulation², as outlined by Bishop [4, pp. 561-563]. This approach involves projecting the data onto a lower-dimensiona linear space, known as the principal subspace, to maximize the variance of the projected data. We will then explore the limitations of PCA, particularly in the context of climate science, and discuss common strategies to address these limitations.

nicrie: make link to Karhunen-Loève expansion

¹Although EOF analysis remains the preferred term in meteorology and climate science, we will use the more commonly recognized term, PCA.

²Hotelling [37] originally proposed this formulation. An alternative derivation, known as the minimum error formulation by Pearson [55], seeks a linear projection that minimizes the average projection error, defined as the mean squared distance between the original data points and their projections.

1.1.1 Derivation from Maximum Variance Formulation

Consider a data set represented as a complex-valued, two-dimensional $N \times D$ matrix $\mathbf{Z}$, where N denotes the number of observations (or samples) and D the number of variables (or features). We will refer to the n^{th} observation as the D -dimensional vector $\mathbf{z}_n$. For simplicity, we assume the data is centered, meaning the mean of each variable is zero, i.e. $\bar{\mathbf{z}} = \frac{1}{N} \sum_{n=1}^N \mathbf{z}_n = \mathbf{0}$. The goal is to project the data onto an M -dimensional space ($M < D$) while maximizing the variance of the projected data.

To start, consider a one-dimensional projection ($M = 1$). We define the direction of this space using a D -dimensional unit³ vector $\mathbf{v}_1$, such that $\mathbf{v}_1^H \mathbf{v}_1 = 1$, where $\mathbf{v}_1^H$ denotes the conjugate transpose of $\mathbf{v}_1$. The data points $\mathbf{z}_n$ are then projected onto this direction, $\mathbf{v}_1^H \mathbf{z}_n$, and the variance of the projected data is

$$\frac{1}{N-1} \sum_{n=1}^{N-1} (\mathbf{v}_1^H \mathbf{z}_n - \mathbf{v}_1^H \bar{\mathbf{z}}) (\mathbf{v}_1^H \mathbf{z}_n - \mathbf{v}_1^H \bar{\mathbf{z}})^H = \mathbf{v}_1^H \mathbf{S}_{ZZ} \mathbf{v}_1, \quad (1.1)$$

where $\mathbf{S}_{ZZ}$ is the covariance matrix of $\mathbf{Z}$ given by

$$\mathbf{S}_{ZZ} = \frac{1}{N-1} \sum_{n=1}^N \mathbf{z}_n \mathbf{z}_n^H = \frac{1}{N-1} \mathbf{Z}^H \mathbf{Z}. \quad (1.2)$$

The goal is to find the vector $\mathbf{v}_1$ that maximizes this variance, formulated as:

$$\underset{\mathbf{v}_1}{\operatorname{argmax}} \mathbf{v}_1^H \mathbf{S}_{ZZ} \mathbf{v}_1 \quad \text{subject to} \quad \mathbf{v}_1^H \mathbf{v}_1 = 1. \quad (1.3)$$

To solve this optimization problem, we maximize the projected variance $\mathbf{v}_1^H \mathbf{S}_{ZZ} \mathbf{v}_1$ with respect to $\mathbf{v}_1$. To enforce the normalization constraint $\mathbf{v}_1^H \mathbf{v}_1 = 1$, a Lagrange multiplier λ_1 is introduced, leading to the unconstrained maximization of:

$$\mathbf{v}_1^H \mathbf{S}_{ZZ} \mathbf{v}_1 + \lambda_1 (1 - \mathbf{v}_1^H \mathbf{v}_1). \quad (1.4)$$

By setting the derivative with respect to $\mathbf{v}_1$ to zero, we find a stationary point when:

³This normalization is for convenience and not necessary, since we are only interested in the direction of $\mathbf{v}_1$, not its magnitude.

$$\mathbf{S}_{ZZ}\mathbf{v}_1 = \lambda_1\mathbf{v}_1. \quad (1.5) \quad 1$$

This indicates that $\mathbf{v}_1$ must be a eigenvector of $\mathbf{S}_{ZZ}$, with λ_1 being the corresponding real-valued⁴ eigenvalue. The variance $\mathbf{v}_1^T \mathbf{S}_{ZZ} \mathbf{v}_1$ reaches its maximum when $\mathbf{v}_1$ aligns with the eigenvector associated with the largest eigenvalue⁵ λ_1 . The eigenvector is referred to as the first **vector of coefficients**⁶ (VC) of the data set. 2 3 4 5

To find additional VCs, we iteratively select new directions that maximize the projected variance while being orthogonal to the previously chosen directions. For an M-dimensional projection space, the optimal linear projection that maximizes the variance of the projected data is defined by the M eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_M$ of the covariance matrix $\mathbf{S}_{ZZ}$, corresponding to the M largest eigenvalues $\lambda_1, \dots, \lambda_M$. This result can be derived using proof by induction, leading to the concise expression: 6 7 8 9 10 11

$$\mathbf{S}_{ZZ}\mathbf{V} = \mathbf{V}\mathbf{\Lambda}, \quad (1.6) \quad 12$$

where $\mathbf{V}$ is unitary, i.e. $\mathbf{V}^T \mathbf{V} = \mathbf{I}_M$. 13

Thus, the eigenvectors of $\mathbf{S}_{ZZ}$ emerge as the new coordiante system "along" regularities in the data. For cases where these regularities are not aligned in unitary directions, see Section 1.1.2. 14 15

According to the spectral theorem for normal matrices [e.g. 36, p. 133], the hermitian covariance matrix $\mathbf{S}_{ZZ}$ can be decomposed into the sum of the outer products of the eigenvectors $\mathbf{v}_i$ and the corresponding real-valued eigenvalues λ_i : 16 17 18

$$\mathbf{S}_{ZZ} = \sum_{i=1}^M \lambda_i \mathbf{v}_i \mathbf{v}_i^H. \quad (1.7) \quad 19$$

Each eigenvalue λ_i represents the amount of variance captured by the corresponding component, typically expressed as a fraction γ_i of the total variance: 20 21

⁴This follows from the property of the covariance matrix being hermitian.

⁵This can easily verified by left-multiplying Equation (1.5) with $\mathbf{v}_1^H$ and making use of the normalization condition.

⁶Naming conventions can vary significantly across different scientific domains. For example, in machine learning, the terms principal components and principal component scores refer to the eigenvectors (here $\mathbf{V}$) and their corresponding data projections (here $\mathbf{P}$) [e.g. 4]. In statistics, these are often called vectors of coefficients/loadings and principal components [e.g. 40], while in climate science and meteorology, they are known as empirical orthogonal functions and principal components/expansion coefficients [e.g. 64]. In this thesis, I aim to maintain consistency with the terminology commonly used in statistics.

$$\gamma_i = \frac{\lambda_i}{\sum_{i=1}^M \lambda_i} \quad (1.8)$$

Projecting the data $\mathbf{Z}$ onto the M -dimensional subspace thus provides a lower dimensional representation, called the principal components (PCs),

$$\mathbf{P} = \mathbf{Z}\mathbf{V}, \quad (1.9)$$

which also enables the reconstruction of the original data from the projected data:

$$\mathbf{Z} = \mathbf{P}\mathbf{V}^H. \quad (1.10)$$

However, calculating the eigenvector decomposition is not always computationally efficient. A more efficient method involves using Singular Value Domposition (SVD) [e.g. 36, p. 150], which decomposes the data matrix as

$$\mathbf{Z} = \mathbf{U}'\mathbf{\Sigma}\mathbf{V}'^H, \quad (1.11)$$

where $\mathbf{U}'$ and $\mathbf{V}'$ are unitary matrices, $\mathbf{U}'^H\mathbf{U}' = \mathbf{V}'^H\mathbf{V}' = \mathbf{I}_M$, and $\mathbf{\Sigma}$ is a diagonal matrix containing the singular values σ_i of $\mathbf{Z}$. The covariance matrix $\mathbf{S}_{ZZ}$ can then be expressed in terms of the SVD as

$$\mathbf{S} = \frac{1}{N-1}\mathbf{Z}^H\mathbf{Z} = \frac{1}{N-1}\mathbf{V}'\mathbf{\Sigma}^H\mathbf{U}'^H\mathbf{U}'\mathbf{\Sigma}\mathbf{V}'^H = \mathbf{V}\mathbf{\Lambda}\mathbf{V}^H, \quad (1.12)$$

where, by comparison with (1.6), $\mathbf{V} = \mathbf{V}'$ and $\mathbf{\Lambda} = \frac{1}{N-1}\mathbf{\Sigma}^T\mathbf{\Sigma}$. This establishes a direct relationship between the eigenvalues of the covariance matrix and the singular values of the data matrix:

$$\lambda_i = \frac{\sigma_i^2}{N-1}. \quad (1.13)$$

Comparing (1.10) with (1.11) shows that the PC scores $\mathbf{P}$ can be normalized

$$\mathbf{U} \equiv \mathbf{U}' = \mathbf{P}\mathbf{\Sigma}^{-1} = \frac{1}{(N-1)^{1/2}} \mathbf{P}\mathbf{\Lambda}^{-1/2}, \quad 1$$

or alternatively $\mathbf{P} = (N-1)^{1/2} \mathbf{U}\mathbf{\Lambda}^{1/2}. \quad (1.14) \quad 2$

Thus, $\mathbf{U}^H \mathbf{U} = \mathbf{I}_M$ ensures that the PC scores $\mathbf{U}$ are unitary and uncorrelated. 3

A common scaling method for the vector of coefficients $\mathbf{V}$ involves scaling by the square root of the corresponding eigenvalues, referred to as **loadings**: 4
5

$$\tilde{\mathbf{V}} = \mathbf{V}\mathbf{\Lambda}^{1/2}. \quad (1.15) \quad 6$$

Although not strictly unitary, these loadings form a generalized unitary matrix, as shown by 7

$$\tilde{\mathbf{V}}^H \tilde{\mathbf{V}} = \mathbf{\Lambda}. \quad (1.16) \quad 8$$

Using this scaling, the Equation 1.10 can be expressed as 9

$$\mathbf{X} = (N-1)^{1/2} \mathbf{U}\tilde{\mathbf{V}}^T. \quad (1.17) \quad 10$$

Computational Considerations The computational cost of performing eigendecomposition on the covariance matrix scales with $O(D^3)$, making it impractical for modern datasets, such as those in climate science, where typical data sizes are on the order of $D = O(10^6)$ or even larger. For example, ERA5 [13], a widely used reference for state-of-the-art reanalysis climate data, distributes a single climate variable on a regular 0.25° grid, resulting in $D = 1\,036\,800$ grid cells. 11
12
13
14
15

In practice, more efficient algorithms, such as SVD or power methods [27], have been employed traditionally employed. Recently, randomized low-rank approximation techniques [22, 30] have further reduced computational complexity to $O(ND \log(M) + (N+D)M^2)$. These advancements offer feasible alternatives for handling large, out-of-memory datasets (see Table 1.1). 16
17
18
19

Shortcomings Despite its widespread use and versatility, PCA has several limitations and potential drawbacks that should be considered: 20
21

- 1 **1. Linearity Assumption:** PCA assumes a linear structure in the data, which may not
2 hold in complex systems such as climate data, where nonlinear relationships can be
3 significant [63].
- 4 **2. Interpretability of VCs:** The orthogonality of the vectors of coefficients is a mathematical
5 construct that may not always correspond to the physical reality of the data, potentially
6 leading to misinterpretations [60].
- 7 **3. Variance Focus:** Focusing on directions of maximum variance may not always capture
8 the most relevant features for a specific application, potentially critical but less variable
9 information [24].
- 10 **4. Static Nature:** As a static method, PCA does not account for temporal dynamics or
11 changes over time, which is particularly limiting in climate science, where time evolution
12 is a key aspect.
- 13 **5. Sensitivity to Outliers:** PCA can be sensitive to outliers, as extreme values can dispro-
14 portionately influence the VCs, leading to misleading results.
- 15 **6. Sensitivity to Scaling:** The results are sensitive to the scaling of the input data, meaning
16 that variables with larger ranges can dominate the VCs unless the data is appropriately
17 standardized.
- 18 **7. Loss of Information:** While dimensionality reduction is often the goal, PCA inherently
19 discards some information, which could lead to the loss of important details crucial for
20 accurate analysis and interpretation in specific applications.

21 Over the past few decades, numerous modifications and extensions have been proposed to
22 address these limitations, especially in the context of climate science. For instance, to address
23 the linearity assumption, various forms of kernel PCA [10, 63, 70] have been developed. For
24 improved interpretability, rotated [60] and sparse [41] PCA methods have been proposed. Ro-
25 bust PCA [16] offers a more resilient approach to handling sensitivity to outlier. Furthermore,
26 extended [8, 69] and complex [1, 34] PCA methods have been introduced to account for tempo-
27 ral dynamics. In addition, eigenmethods like Optimal Persistence Analysis (OPA) [19, 20] and
28 Principal Oscillation Pattern (POP) analysis [32] provide alternative approaches independent
29 to variance maximization to understanding dominant patterns in physical climate fields.

30 In the following sections, we will explore some of these shortcomings and proposed solutions
31 in greater detail. Specifically, we will discuss how additional rotation of the PCA solution can
32 enhance physical interpretability (Section 1.1.2) and how complexifying the data can provide
33 insights into time evolution patterns (Section 1.1.3).

Table 1.1 Computational costs of different algorithms used to compute either full or approximative M -rank solutions for PCA for a real, dense, in-memory matrix with dimensions $N \times D$. Compiled from Bishop [4] and Halko *et al.* [30].

Algorithm	Complexity
Eigenvector decomposition	$O(D^3)$
Full SVD	$O(ND \min(N, D))$
Power method	$O(MD^2)$
Truncated SVD	$O(NDM)$
Randomized SVD	$O(ND \log(M) + (N + D)M^2)$

1.1.2 Sparse PCA via Rotation

PCA is widely used for dimensionality reduction, as it maximizes the explained variance and provides orthogonality, making it optimal in a least-squares sense [4]. It effectively identifies principal axes that capture the covariance among variables, allowing for the examination of interrelationships between stations, grid points or variables. However, PCA is not ideal for identifying the cause of variation or deriving physical modes of variability [40, 45, 60].

One limitation is that real-world processes do not necessarily adhere to the orthogonality imposed by PCA's eigenvectors. Additionally, the dense nature of these VCs – where all elements are non-zero – can make interpretation challenging. In the context of climate and environmental science, where variables are often represented across spatial grids, these limitations can lead to recognizable but artificial patterns known as Buell patterns [9, 44, 60]. These patterns typically start with a positive component and evolve into increasingly complex structures, such as bipole or tripole patterns (Figure 1.1). Such issues highlight the disconnect between standard PCA and underlying physical processes [60], prompting the development of alternative methods that produce sparse VCs, characterized by many zero or near-zero elements. This sparsity – which necessarily implies the loss of orthogonality for at least one of the VCs and PC scores – can enhance interpretability by localizing the patterns and reducing the density of the VCs.

In climate science, the most common approach to achieving sparse PCA is through rotation techniques [35, 59, 60], a concept borrowed from factor analysis [cf. 40, pp. 269]. In this "factor analysis viewpoint" of PCA [40, pp. 150-161], a linear transformation is applied to a subset of the VCs to derive a new set of rotated VCs that are "simpler" in a heuristically defined manner, ideally aligning better with the underlying physical processes.

As noted in Equation (1.17), once the orthogonality constraint is relaxed, PCA solutions exhibit inherent indeterminacy, allowing any non-singular transformation $\mathbf{R}$ to satisfy the PCA formulation [e.g. 60, p. 305]:

$$\mathbf{Z} = (\mathbf{N} - 1)^{1/2} \mathbf{U} \tilde{\mathbf{V}}^H = (\mathbf{N} - 1)^{1/2} \mathbf{U} \mathbf{R} \mathbf{R}^{-1} \tilde{\mathbf{V}}^H. \quad (1.18)$$

Rotated PCA exploits this indeterminacy by choosing $\mathbf{R}$ to optimize a simplicity criterion f , resulting in rotated loadings⁷ $\tilde{\mathbf{V}}_R = \tilde{\mathbf{V}} \mathbf{R}$ that maximize f :

$$\underset{\mathbf{R}}{\operatorname{argmax}} f(\tilde{\mathbf{V}}_R). \quad (1.19)$$

There are two main categories of rotation: *orthogonal*⁸ and *oblique*. In orthogonal rotation, $\mathbf{R}$ is a unitary matrix, preserving the uncorrelated nature of the PCs since $\mathbf{U}_R^H \mathbf{U}_R = \mathbf{R}^H \mathbf{U}^H \mathbf{U} \mathbf{R} = \mathbf{I}_M$. In oblique rotation, $\mathbf{R}$ is a general non-singular matrix, resulting in correlated PCs, as $\mathbf{U}_R^H \mathbf{U}_R = \mathbf{R}^H \mathbf{R}$. Regardless of the method, the rotated loadings are not orthogonal,

$$\tilde{\mathbf{V}}_R^H \tilde{\mathbf{V}}_R = \mathbf{R}^H \tilde{\mathbf{V}}^H \tilde{\mathbf{V}} \mathbf{R} \stackrel{(1.16)}{=} \mathbf{R}^H \mathbf{\Lambda} \mathbf{R} \neq \mathbf{I}_M, \quad (1.20)$$

which is crucial for achieving a more interpretable solution. Various simplicity criteria f have been proposed [see e.g. 60], with most focusing on making the rotated loadings more localized and physically interpretable. In practice, the specific choice of simplicity criterion often has little impact on the results [60].

Varimax Orthogonal Rotation The most widely used orthogonal rotation technique is the varimax criterion [42], which aims to simplify the VCs by maximizing the variance of the squared loadings,

$$f(\tilde{\mathbf{V}}_R) = \sum_{m=1}^M \left(p \sum_{j=1}^P b_{jm}^2 - \left(\sum_{j=1}^P b_{jm} \right)^2 \right) \quad (1.21)$$

⁷In principle, it is possible to rotate the unitary matrix of coefficients $\mathbf{V}$ instead of the loadings matrix $\tilde{\mathbf{V}}$ [40, 51]. However, this approach is seldom used in practice. The primary reason is that rotating the unitary coefficient matrix preserves the orthogonality of the VCs but results in the loss of orthogonality among the PCs, which does not address issues like the Buell patterns. Moreover, this method does not consider the relative importance of the spatial patterns, as indicated by the eigenvalues, making the results of the rotation process highly sensitive to the selection of the number of components to rotate.

⁸In the broader context of complex data, this would be referred to as a unitary rotation. However, since factor analysis has primarily been applied to real-valued data, the terms orthogonal and oblique rotations have become the standard. To avoid confusion, I use these historical terms while acknowledging that they also encompass the more general case of complex data.

1.1 Principal Component Analysis

9

where b_{jm} are the squared norm of the rotated loadings $|\tilde{v}_{R,jm}|^2/h_j^2$ normalized by the communalities $h_j^2 = \sum_{m=1}^M |\tilde{v}_{R,jm}|^2$ of variable j . The varimax rotation pushes the elements of $\tilde{\mathbf{V}}_R$ towards either 0 or 1, creating a sparse solution with primarily large or small values, which is particularly useful for identifying localized physical processes, such as atmospheric or oceanic teleconnections.

Promax Oblique Rotation The promax criterion [33], a generalization of the varimax criterion, is one of the most common oblique rotation techniques. It is based on the premise that an orthogonal simple structure solution, such as varimax, is often close to an optimal oblique solution. The promax method constructs a hypothetical oblique target matrix $\mathbf{T}$ by performing an oblique Procrustes transformation [38] on the varimax-rotated loadings matrix,

$$\mathbf{T} = \tilde{\mathbf{V}}_R \mathbf{R} + \mathbf{E}, \quad (1.22)$$

where $\mathbf{R}$ is the transformation matrix and $\mathbf{E}$ is a matrix of residuals. The normal equation for solving the complex Procrustes problem is given by:

$$\mathbf{R} = (\tilde{\mathbf{V}}_R^H \tilde{\mathbf{V}}_R)^{-1} \mathbf{T}^H \tilde{\mathbf{V}}_R. \quad (1.23)$$

To define the target matrix $\mathbf{T}$, the promax criterion increases the discrimination between high and low loadings by raising the loadings to a power q while retaining the original sign,

$$t_{jm} = \frac{|a_{jm}|^{q+1}}{a_{jm}}, \quad (1.24)$$

where q controls the degree of obliqueness and a_{jm} are the elements of the max-normalized, varimax-rotated loadings matrix $\mathbf{A}$, defined as

$$\mathbf{A} = \tilde{\mathbf{A}} \mathbf{K}^{-1}, \quad (1.25)$$

$$\tilde{\mathbf{A}} = \mathbf{H}^{-1} \tilde{\mathbf{V}}_R, \quad (1.26)$$

with $\mathbf{K}$ being a diagonal matrix containing the maximum absolute values of each column of $\tilde{\mathbf{A}}$, $a_m = \max_m |A_{pm}|$, and $\mathbf{H}$ a diagonal matrix with the square root of the communalities h_j for the j^{th} variable.

Application and Criticism Both orthogonal and oblique rotation techniques, particularly varimax and promax, have been extensively used since the 1980s to identify stationary modes of variability in climate data [e.g. 2, 26, 35, 39, 43, 48, 52]. Notable climate oscillations, such as the North Atlantic Oscillation (NAO), have been defined based on varimax-rotated patterns of loadings [2]. These techniques also help identify "ghost patterns", artificial patterns identified by PCA that are not physically meaningful [39, 47].

Rotated modes offer several advantages, such as greater robustness against temporal and spatial subsampling [17] compared to unrotated modes. They also effectively mitigate the statistical artifacts associated with Buell patterns, resulting in more realistic representations of the underlying autocorrelation structure and thus increased interpretability [47, 60, 61]. Moreover, rotation techniques have been successfully integrated into other PCA extensions dealing with the non-linear and non-stationary nature of climate oscillations [10, 29, 34].

However, rotation techniques are not without criticism, mainly due to the ad-hoc nature of the various simplicity criteria [cf. 40, pp. 270-274]. This has led to the exploration of alternative approaches to enforce sparsity. Notably, Jolliffe *et al.* [41] introduced the Simplified Component Technique - LASSO (SCoTLASS), a sparse PCA variant based on the least absolute shrinkage and selection operator (LASSO), originally from regularized regression [65]. This method imposes a constraint on the ℓ_1 norm of the vector of loadings, resulting in some loading coefficients being exactly zero.

Despite its advantages, SCoTLASS has drawbacks, particularly when the number of variables D is much larger than the number of observations N . In such cases, SCoTLASS can only select at most N non-zero elements in the loading vectors, which is not ideal in climate science where often $D \gg N$. Additionally, since SCoTLASS solutions cannot be obtained via SVD and require a numerical approach using projected gradient methods [66], the computational demand is typically higher than for (rotated) PCA. While this method has inspired many other sparse PCA techniques based on penalties or constraints (usually ℓ_0 or ℓ_1) [see e.g. 23, 73, 74], these formulations are often complex and require algorithmic innovations to avoid issues like local optima and ensure convergence. They may also face challenges with memory limitations, slow computation speeds and limited effectiveness in modelling the data [15, 28].

It is likely for these reasons, that rotated PCA remains one of the most commonly used techniques for achieving sparse PCA solutions in climate science. It is computationally efficient, straightforward to implement and provides a more interpretable understanding of the data compared to standard PCA.

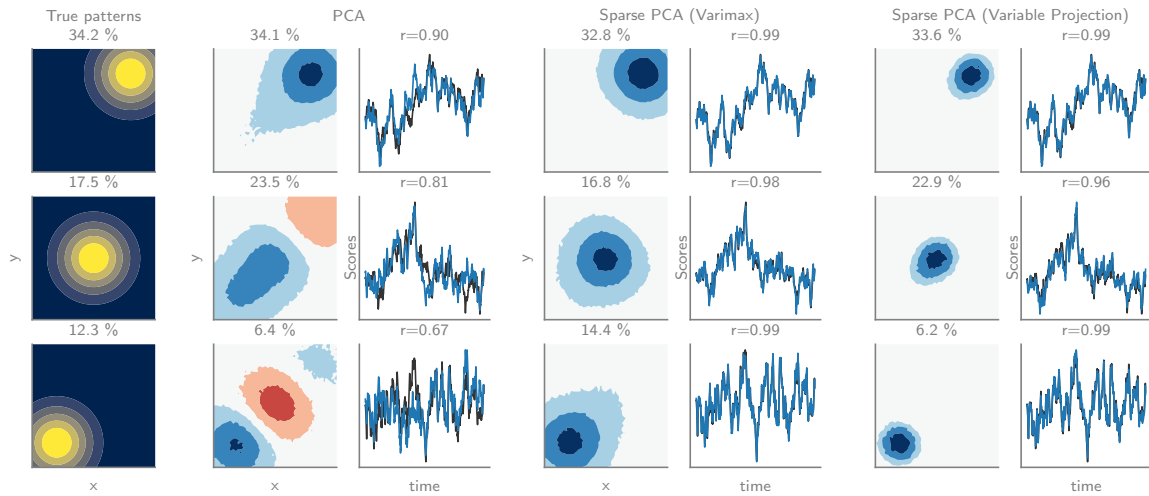


Fig. 1.1 Comparison of the first three modes obtained from PCA vs. Varimax-rotated PCA for a synthetic data set. (Left) Original data set with three different, uncorrelated autoregressive processes. (Middle) PCA solution. (Right) Varimax-rotated PCA solution.

1.1.3 Hilbert PCA

While PCA efficiently identifies the principal axes along which the covariance among all variables is distributed, and rotated PCA provides a more sparse solution related to underlying physical processes, both methods are inherently static. They do not account for temporal dynamics or changes over time, as they only consider the current state of the physical field. This limitation is significant in climate science, where time evolution is crucial. For instance, standard PCA does not differentiate between the order of data points over time, resulting in the same eigenvectors and eigenvalues regardless of temporal permutations.

This static nature poses a challenge for capturing oscillatory patterns, such as QBO, MJO, ENSO and BSISO, which involve spatial patterns propagating over time. These phenomena cannot be adequately captured by standard PCA.

Hilbert Transform as a Momentum Operator

To address this limitation, one approach is to model both the current state of the climate and its tendency or 'momentum', analogous to the Hamiltonian principle in mechanics, where a system's future is described by first-order differential equations for location (state) and momentum (tendency) [cf. 64, p. 353]. In classical mechanics, propagating features are represented using canonical coordinates: each state variable is augmented by a momentum variable, allowing oscillations to be depicted as rotations in a two-dimensional phase space.

From a data perspective, this augmentation can be achieved via the Hilbert transform ($\mathcal{H}$). For a real time series $x(t)$, the Hilbert transform $x'(t)$ is defined as

$$x'(t) \equiv \mathcal{H}\{x\}(t) = \frac{1}{\pi} \text{p.v.} \int_{-\infty}^{\infty} \frac{x(\tau)}{\tau - t} d\tau, \quad (1.27)$$

where p.v. denotes the Cauchy principal value ensuring convergence despite the singularity at $\tau = t$. The Hilbert transform emphasises the local properties of $x(t)$ by convolution with $\frac{1}{\pi t}$. In the frequency domain, it is related to the Fourier transform $\mathcal{F}$ by

$$\mathcal{F}\{y\}(\omega) = -i \operatorname{sgn}(\omega) \mathcal{F}\{x\}(\omega) \quad (1.28)$$

where the imaginary unit i indicates a phase shift of $\pi/2$ for the frequency components ω of x , placing x' in quadrature to x . This provides an intuitive interpretation of the Hilbert transform, as it acts as a linear filter that alters the phase of the signal's components by 90 degrees without changing their amplitude.

By using the Hilbert transform, a real-valued time series $x(t)$ can be complexified into $z(t) = x(t) + i\mathcal{H}\{x\}(t)$, known as the *analytic signal*, which allows for the computation of the instantaneous amplitude and phase of $x(t)$:

$$A(t) = |z(t)| = \sqrt{x(t)^2 + \mathcal{H}\{x\}(t)^2}, \quad \phi(t) = \arg(z(t)) = \arctan\left(\frac{\mathcal{H}\{x\}(t)}{x(t)}\right). \quad (1.29)$$

Similar results can be obtained for the discrete case [54, pp. 942-956], and this concept can be extended to multivariate data [e.g. 31, pp. 105-106], defining the analytic signal of a data matrix X as

$$Z = X + i\mathcal{H}\{X\}, \quad (1.30)$$

where $\mathcal{H}\{X\}$ is the feature-wise Hilbert transform of X .

Rasmusson *et al.* [57] introduced the concept of Hilbert PCA using this data augmentation technique⁹. This method, refined by Barnett [1] and Horel [34], involves applying PCA to the augmented data matrix Z . By incorporating phase information, Hilbert PCA can capture

⁹Although Rasmusson *et al.* [57] originally termed it *Hilbert Singular Decomposition*, the terms *Hilbert EOF/PCA* and *Complex EOF analysis* are more commonly used in the literature [12, 31, 40, 64].

the direction and speed of wave-like patterns, unlike traditional PCA, which only identifies standing oscillations. The resulting eigenvectors and principal component scores are complex, with the real and imaginary components linked via the Hilbert transform. The variance capture by each component is represented by the real eigenvalues of the complex covariance matrix of Z (see Section 1.1).

The main advantage of Hilbert PCA is the ability to identify and reconstruct propagating waves, which is especially valuable in studying climate phenomena characterized by such patterns, including MJO, ENSO, QBO and BSISO. For instance, Cai & Baines [14] utilized Hilbert PCA to analyze SST in the southern hemisphere, linking the findings to ENSO, while Bürger [12] employed it to develop an index for BSISO. However, to reliably identify oscillatory patterns, sufficiently long time series are required to resolve at least one full oscillation [50]. Additionally, interpreting of the Hilbert transform as representing the instantaneous frequency is valid holds for narrow-band signals; the phase of the Hilbert transform is not well-defined for broadband signals [6, 25]. Moreover, real-time monitoring is challenging with Hilbert PCA because new real-valued data cannot be directly projected onto the existing patterns without knowing or estimating the exact momentum (phase) of the new data, which requires future information (Equation (1.27)). To mitigate this, an approximative real-time (causal) Hilbert transform can be employed [11, 12], offering a practical remedy to the issue of real-time applicability. Finally, Hilbert PCA shares PCA's limitations regarding physical interpretability due to the orthogonality of the VCs. To address this, rotated Hilbert PCA has been proposed to enhance interpretability [5, 35].

1.2 | Maximum Covariance Analysis (MCA)

Sometimes we're not so much interested in the primary modes of variations within a dataset, but rather the modes of shared variations between two different data sets. Consider the MJO, which is the largest element of the intraseasonal variability in the tropical atmosphere, characterized by an eastward migrating cell of overturning wind circulation within the troposphere that lead to large regions of enhanced and suppressed precipitation. Rather than interested in the dominant variability in any of the individual fields, we may want to find the shared dynamics between these variables ignoring dominant variability that is associated only to one of the variables such as diurnal cycles linked to surface processes but which are absent in higher levels of the atmosphere.

Maximum Covariance Analysis (MCA) is a statistical method that has exactly this goal: to identify patterns of maximum covariance between two (or more) different datasets [64, p. 321]. It is a data-driven approach similar to PCA, in the sense that we can formulate it as optimization problem where we want to find pairs of coupled (complex) patterns $v_{i,1}$ for two

- 1 (complex-valued) datasets, $Z_i, i = [x, y]$, such that the projection of the data onto these patterns
 2 maximizes the covariance between the two datasets,

$$3 \quad \underset{\mathbf{v}_{x,i}}{\operatorname{argmax}} = (\mathbf{v}_{x,1} \mathbf{C}_{xy} \mathbf{v}_{y,1})^2 \quad \text{subject to} \quad \|\mathbf{v}_{x,i}\|_2 = 1, \quad (1.31)$$

- 4 where the cross-covariance matrix between the two datasets is given by

$$5 \quad \mathbf{C}_{xy} = \frac{1}{N-1} \mathbf{Z}_x^H \mathbf{Z}_y. \quad (1.32)$$

- 6 The solution to this problem gives rise to a generalized eigenvalue equations similar to that of
 7 PCA, which can also be solved by using the SVD of the cross-covariance matrix

$$8 \quad \mathbf{C}_{xy} = \mathbf{V}_x \mathbf{\Sigma} \mathbf{V}_y^H \quad (1.33)$$

- 9 where the m^{th} column of the left and right singular vectors, $\mathbf{V}_x$ and $\mathbf{V}_y$, contains the patterns
 10 $\mathbf{v}_{x,m}$ and $\mathbf{v}_{y,m}$, and $\mathbf{\Sigma}$ a diagonal matrix containing the singular values σ_i in decending order.
 11 Then it can be easily verified that the first left and right singular vectors corresponding to the
 12 first column maximize Equation (1.31), resulting in a corresponding maximum covariance
 13 given by the largest singular value σ_1 .

- 14 Since the patterns are unitary, we can expand the datasets according to

$$15 \quad \mathbf{Z}_i = \mathbf{P}_i \mathbf{V}_i^H, \quad (1.34)$$

- 16 with the expansion coefficient $\mathbf{P}_i$. The expansion coefficients $\mathbf{P}_1$ and $\mathbf{P}_2$ are uncorrelated with
 17 each other, since

$$18 \quad \mathbf{P}_1^H \mathbf{P}_2 = \mathbf{V}_1^H \mathbf{Z}_1^H \mathbf{Z}_2 \mathbf{V}_2 \stackrel{(1.32)}{=} (N-1) \mathbf{V}_1^H \mathbf{C}_{xy} \mathbf{V}_2 \stackrel{(1.33)}{=} (N-1) \mathbf{\Sigma}. \quad (1.35)$$

- 19 In contrast, the expansion coefficients $\mathbf{P}_1$ and $\mathbf{P}_2$ are, in general, correlated with themselves, i.e.
 20 $\mathbf{P}_i^H \mathbf{P}_i \neq c \mathbf{I}_m$ for any arbitrary constant c .

- 21 The total covariance is given by squared Frobenius norm of the cross-covariance matrix,

$$\|C_{xy}\|_F^2 = \sum_{i=1}^M \sigma_i^2, \quad (1.36) \quad 1$$

which shows that the i^{th} mode captures the squared covariance σ_i^2 , thus its relative importance can be expressed as the squared covariance fraction $f_{xy,i}$: 2
3

$$f_{xy,i} = \frac{\sigma_i^2}{\sum_{i=1}^M \sigma_i^2}. \quad (1.37) \quad 4$$

This provides a measure of the relative importance of the i^{th} mode in capturing the shared variability between the two datasets. However, to quantify the importance of the patterns themselves, we need to consider the variance within the individual datasets explained by each mode in relation to the total variance of the individual datasets, which can be expressed as the squared Frobenius norm of the individual datasets, 5
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$$f_{xx,i} = \frac{\mathbf{v}_{x,i}^H \mathbf{C}_{xx} \mathbf{v}_{x,i}}{\|\mathbf{Z}_x\|_F^2} = \frac{1}{N-1} \frac{\mathbf{p}_{x,i}^H \mathbf{p}_{x,i}}{\|\mathbf{Z}_x\|_F^2}, \quad (1.38) \quad 10$$

$$f_{yy,i} = \frac{\mathbf{v}_{y,i}^H \mathbf{C}_{yy} \mathbf{v}_{y,i}}{\|\mathbf{Z}_y\|_F^2} = \frac{1}{N-1} \frac{\mathbf{p}_{y,i}^H \mathbf{p}_{y,i}}{\|\mathbf{Z}_y\|_F^2}. \quad (1.39) \quad 11$$

The method was popularized in the field of meteorology¹⁰ by Bretherton *et al.* [7] and Wallace *et al.* [68] since when it has been extensively applied [e.g. 3, 46, 53, 58, 67, 68] 12
13

Why Is It Important? Understanding Coupled Variability: MCA is crucial for understanding the coupled variability between different climatic variables. For instance, it can help identify how changes in sea surface temperatures may be related to atmospheric pressure patterns, which is vital for understanding phenomena like El Niño-Southern Oscillation (ENSO). 14
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Teleconnections Analysis: It is used to study teleconnections, which are climate anomalies related to each other at large distances (thousands of kilometers apart). By identifying the dominant modes of variability, MCA helps in understanding how different regions of the climate system interact. 18
19
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21

¹⁰According to Jolliffe [40, p. 225], it was first used in psychology starting from the 1950s where it is known as inter-battery factor analysis. As a variant of Partial Least Squares (PLS), it was later used in biomedical sciences by Sampson *et al.* [62] and termed PLS-SVD by Wegelin [71]- In meteorology, Bretherton *et al.* [7] coined the confusing term SVD analysis, which is still in use today [e.g. 53]. Rarely, the method is also called Canonical Covariance Analysis [e.g. 18, 31], akin to Canonical Correlation Analysis.

1 Climate Prediction: The patterns identified by MCA can be used for predictive modeling. By
2 understanding the leading modes of variability, scientists can make better predictions about
3 future climate conditions.

4 Data Reduction and Interpretation: MCA helps in reducing the dimensionality of large climate
5 datasets by focusing on the most significant modes of variability, making it easier to interpret
6 complex interactions.

7 Whereas PCA works with a single set of random variables, MCA considers two (or more)
8 variables and tries to find a corresponding pair of linear subspaces that have high cross-
9 covariance, so that each component within one of the subspaces exhibits high covariance
10 with a single component from the other subspace. Its solution can be expressed in terms of a
11 generalized eigenvector problem.

12 The origins of Maximum Covariance Analysis (MCA) date back to Tucker (1958), where it is
13 known as inter-battery factor analysis; The method was popularised in atmospheric science by
14 Bretherton et al. (1992) and Wallace et al. (1992) under the name singular value decomposition
15 (SVD) analysis. the name maximum covariance analysis is used by von Storch and Zwiers
16 (1999, Section 14.1.7)

17 MCA reduces to PCA when both sets are the same, but there's an even deeper link to PCA:
18 maximum covariance analysis can be viewed as two PCAs, followed by rotation to match up
19 the results of the two analyses as closely as possible (Cherry, 1997).

20 difference to CCA: Maximum covariance analysis differs from CCA in two ways: covariance
21 rather than correlation is maximized, and vectors of loadings are orthogonal instead of derived
22 variates being uncorrelated. The rationale behind maximum covariance analysis is that it may
23 be important to explain a large proportion of the variance in one set of variables using the
24 other set, and a pair of variates from CCA with large correlation need not necessarily have
25 large variance

26 Wilks: There are two main distinctions between CCA and MCA. The first is that CCA maxi-
27 mizes correlation, whereas MCA maximizes covariance. The leading CCA modes may capture
28 relatively little of the corresponding variances (and thus yield small covariances even if the
29 canonical correlations are high). On the other hand, MCA will find linear combinations with
30 large covariances, which may result more from large variances than a large correlation. The
31 second difference is that the vectors a_m and b_m in maximum covariance analysis are orthogo-
32 nal, and the projections v_m and w_m of the data onto them are in general correlated, whereas
33 the canonical variates in CCA are uncorrelated but the corresponding canonical vectors are not
34 generally orthogonal. Bretherton et al. (1992), Cherry (1996), Tippett et al. (2008), and Van den
35 Dool (2007) compare the two methods in greater detail.

1.3 Canonical Correlation Analysis

17

Prediction via

$$\mathbf{P}_1 = \mathbf{X}_1 \mathbf{V}_1 \quad (1.40)$$

$$\mathbf{P}_2 = \mathbf{X}_2 \mathbf{V}_2 \quad \text{or} \quad \mathbf{P}_2 = \mathbf{P}_1 \mathbf{G} \quad \text{with} \quad g_k = \frac{s_k}{\text{varp}_{1,k}} \quad (1.41)$$

$$\mathbf{X}_2 = \mathbf{P}_2 \mathbf{V}_2^T = \mathbf{P}_1 \mathbf{G} \mathbf{V}_2^T = \mathbf{X}_1 \mathbf{V}_1 \mathbf{G} \mathbf{V}_2^T \quad (1.42)$$

Accuracy of reconstruction depends on

- The degree of covariance between the two fields.
- The cumulative squared covariance fraction represented by the leading modes. The more modes the better the reconstruction.
- The variance explained by the right singular vectors in the right phase space. I.e. the component of the right space that is related to the left space.

1.2.1 Rotated Maximum Covariance Analysis

1.2.2 Complex Maximum Covariance Analysis

CCA or MCA?

MCA focuses on covariance: Modes tend to be large where the variance is large. Danger that physical modes are confounded by large variance.

CCA focuses on correlation: Coupling is identified also if associated variance is low. Danger that physical modes are confounded by small (insignificant) variations (sampling problems)

1.3 | Canonical Correlation Analysis

Finally, it is worth noting that there exists a closely related linear dimensionality reduction technique called canonical correlation analysis, or CCA (Hotelling, 1936; Bach and Jordan, 2002). Whereas PCA works with a single random variable, CCA considers two (or more) variables and tries to find a corresponding pair of linear subspaces that have high cross-correlation, so that each component within one of the subspaces is correlated with a single component from the other subspace. Its solution can be expressed in terms of a generalized eigenvector problem.

1.3.1 Complex Canonical Correlation Analysis

1.4 | Neural Networks

1.4.1 Introduction to Neural Networks

Definition and Overview: Explain what neural networks are, including a brief history and their importance in machine learning. Basic Architecture: Describe the basic structure of a neural network, including neurons, layers (input, hidden, output), activation functions, and the learning process.

1.4.2 Recurrent Neural Networks

Introduction to RNNs¹¹: Describe the architecture of RNNs, their ability to handle sequential data, and their applications. Challenges with RNNs: Explain issues such as vanishing and exploding gradients.

Long Short-Term Memory (LSTM) Networks

Introduction to LSTM¹²: Discuss the motivation behind LSTM networks and how they address the limitations of standard RNNs. LSTM Architecture: Describe the components of LSTM cells (input gate, forget gate, output gate) and their functions. Mathematical Formulation: Provide the equations governing LSTM networks. Applications of LSTM: Highlight key applications where LSTMs have shown significant performance improvements.

1.4.3 Explainable AI Techniques

Introduction to Explainable AI: Explain the need for explainable AI, particularly in critical applications where understanding model decisions is crucial. Overview of Techniques: Briefly list various explainable AI methods, with a focus on LRP and SHAP.

Layerwise Relevance Propagation (LRP)

Concept of LRP¹³: Describe what LRP is and its importance in understanding neural network decisions. Mechanism of LRP: Explain how LRP works, including the propagation rules and

¹¹Graves 2021 Supervised Sequence Labelling with Recurrent Neural Networks

¹²Hochreiter & Schmidhuber 1997

¹³Bach et al. 2015

the relevance score computation. Applications of LRP: Provide examples of how LRP is used in practice to interpret model predictions.

simplest continuous latent variable model assumes Gaussian distributions for both the latent and observed variables. simplest continuous latent variable model assumes Gaussian distributions for both the latent and observed variables

SHapley Additive exPlanations (SHAP)

Introduction to SHAP¹⁴: Discuss the SHAP method, its foundation in cooperative game theory, and its advantages. SHAP Values: Explain how SHAP values are computed and their interpretation. Applications of SHAP: Provide examples of SHAP in various domains, highlighting its ability to provide consistent and locally accurate explanations.

1.5 | Gaussian Processes

Class of pattern recognition techniques that typically require a metric to be defined that measures the similarity of any two vectors in input space, and are generally fast to "train" but slow at making predictions for test data points.

1.5.1 Introduction to Gaussian Processes

Definition and Overview: Explain what Gaussian Processes are, their significance in machine learning, and their use in modeling continuous data. Basic Concepts: Introduce key concepts such as Gaussian distributions, covariance functions, and kernels.

1.5.2 Gaussian Processes for Regression

GP Regression Model: Describe the formulation of GP for regression tasks, including prior, likelihood, and posterior distributions. Kernel Functions: Discuss common kernel functions (e.g., squared exponential, Matérn) and their role in defining the GP's properties. Hyperparameter Tuning: Explain how to tune hyperparameters using techniques like maximum likelihood estimation.

¹⁴Lundberg 2017

1.5.3 Spatial Modeling with Gaussian Processes

Spatial Data: Define spatial data and the importance of modeling spatial dependencies. GPs in Spatial Modeling: Explain how GPs are particularly suited for spatial data due to their ability to model complex, non-linear relationships and provide uncertainty estimates.

Log Gaussian Cox Process

Introduction to Cox Processes: Explain Cox processes and their relevance in spatial point process modeling. Log Gaussian Cox Model: Describe the log Gaussian Cox model, which assumes the log intensity function is modeled by a GP. Mathematical Formulation: Provide the mathematical formulation of the log Gaussian Cox model.

Negative Binomial Distribution

Introduction to Negative Binomial Distribution: Explain the negative binomial distribution and its use in modeling overdispersed count data. Integration with Log Gaussian Cox Model: Describe how the negative binomial distribution can be integrated into the log Gaussian Cox model to handle count data with overdispersion.

1.5.4 Model Implementation and Inference

Model Fitting: Discuss methods for fitting the log Gaussian Cox model with a negative binomial distribution, including Markov Chain Monte Carlo (MCMC) and variational inference. Inference and Prediction: Explain how to perform inference and make predictions using the fitted model.

1.6 | Comparative Perspective

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Proofs

The Frobenius norm $\|\cdot\|_F$ of a complex $N \times P$ matrix $\mathbf{A} \in \mathbb{C}^{N \times P}$ is defined as:

$$\|\mathbf{A}\|_F \equiv \left(\sum_i^N \sum_j^P |\alpha_{ij}|^2 \right)^{1/2} = \left(\sum_i^{\min(N,P)} \sigma_i^2(\mathbf{A}) \right)^{1/2}. \quad (\text{A.1})$$

This expression reveals that the Frobenius norm is equivalent to the ℓ_2 -norm of the matrix's singular values σ_i . We can demonstrate this equivalence using the singular value decomposition (SVD) of $\mathbf{A}$ and leveraging the property that the trace of a matrix remains invariant under cyclic permutations.

Starting with the definition of the Frobenius norm, we have:

$$\|\mathbf{A}\|_F = \left(\text{tr}(\mathbf{A}^H \mathbf{A}) \right)^{1/2}$$

Applying the SVD, we can express $\mathbf{A} = \mathbf{V} \mathbf{\Sigma} \mathbf{V}^H$, leading to:

$$\|\mathbf{A}\|_F = \left(\text{tr}(\mathbf{V} \mathbf{\Sigma} \mathbf{U}^H \mathbf{U} \mathbf{\Sigma}^H \mathbf{V}^H) \right)^{1/2}$$

Since $\mathbf{U}^H \mathbf{U} = \mathbf{I}$, this simplifies to:

$$\|\mathbf{A}\|_F = \left(\text{tr}(\mathbf{V} \mathbf{\Sigma}^2 \mathbf{V}^H) \right)^{1/2}$$

Finally using the invariance property of the trace, we obtain:

$$\|\mathbf{A}\|_F = \left(\sum_i^{\min(N,P)} \sigma_i^2(\mathbf{A}) \right)^{1/2}$$

This concludes the proof that the Frobenius norm of a matrix is indeed the ℓ_2 -norm of its singular values.

Derivation of Maximum Covariance Analysis

Here we derive the Maximum Covariance Analysis (MCA) from the framework of Generalized Canonical Correlation Analysis where we maximize covariance instead of correlation. The objective function of MCA is:

$$\operatorname{argmax}_{\mathbf{v}_x, \mathbf{v}_y} \|\mathbf{v}_x^H \mathbf{X} \mathbf{Y}^H \mathbf{v}_y\|^2$$

which is a generalization of the Canonical Correlation Analysis (CCA) algorithm. We start by defining the covariance matrix of two random variables $\mathbf{x}$ and $\mathbf{y}$ as:

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